

Pre-Week Homework 05

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1. If n is an integer then $n^2 + 3n + 5$ is odd.
Case 1: If n is even $\rightarrow n = 2k$
 $\rightarrow (2k)^2 + 3(2k) + 5 \Rightarrow 4k^2 + 6k + 5$
 $\Rightarrow (4k^2 + 6k + 4) + 1$
 $\Rightarrow 2(4k^2 + 3k + 2) + 1$
 $\Rightarrow \boxed{2m + 1}$
Case 2: If n is odd $\rightarrow n = 2k + 1$
2. If $3 \mid x$ or $3 \mid y$ then $3 \mid xy$.
Case 1: If some $a \in \mathbb{Z}$, then $x = 3a$
 $\rightarrow xy = (3a)y$
 $\Rightarrow xy = 3(ay)$
 $\Rightarrow \boxed{xy = 3k}$
Case 2: If some $b \in \mathbb{Z}$, then $y = 3b$
3. If two integers have the same parity, their sum is even.
4. If two integers have opposite parity, their product is even.